ASSIGNMENT – 0 (COURSE INSIGHTS)

Name: Tejas Pawar

Roll No: 24102B0033

Batch: CMPN _ B

Subject: E-waste & Environmental Management (EWEM)

Semester: V

Q1. Curriculum Structure

1a. How many buckets are there in the VIT autonomous curriculum?

Our VIT autonomous curriculum is structured into 7 distinct buckets according to the Academic Administration Plan (AAP):

- BSC/ESC (Basic Science & Engineering Science Courses)
- Program Courses (PC)
- Multidisciplinary Courses (MDC)
- Skill Courses
- Experiential Learning (ELC)
- HSSM (Humanities, Social Sciences & Management)
- Liberal Learning

1b. In which bucket is this course included?

This subject is classified under the **HSSM** bucket as a Value Education Course (VEC02).

1c. How many credits need to be completed from this bucket?

The exact total credit requirement for the entire HSSM bucket is not explicitly specified in the course overview documents.

1d. How many credits from this bucket have you completed till now?

0 credits completed so far, as we are just beginning this semester.

1e. How many credits does this course have?

This is a **2-credit** course.

Q2. Prerequisites & Preparation

2a. Have you gone through the curriculum of your course?

Yes, I have thoroughly reviewed the provided Academic Administration Plan (AAP) and syllabus.

2b. Is there a pre-requisite of this course?

Yes. The AAP explicitly lists **"Environmental science"** (studied in the 12th grade or earlier semesters) as a foundational prerequisite for this course.

2c. If yes, do you clearly remember the concepts?

I have a basic understanding and recall the fundamental concepts of environmental studies and pollution.

2d. Preparations required for this course:

To prepare effectively, I need to:

- Brush up on basic environmental science concepts.
- Familiarize myself with current e-waste pollution and recycling strategies.
- Since certain assignments (like the Data Analysis Activity) require visualizing global e-waste datasets, I should ensure my technical skills in Python, Excel, or Tableau are sharp.
- No specific heavy software installation is mandatory right away.

2e. Future course(s) that build on this course?

This subject acts as a stepping stone for upcoming electives in Semester 7 or 8, specifically: *Environmental Management* and *Sustainability Management*.

Q3. Course Outcomes (COs) & Program Outcomes (POs)

3a. List all Course Outcomes (COs). Which CO is most useful?

- **CO1:** Understand need and concept of e-waste management & recycling.
- **CO2:** To gain the knowledge of legislative rules and regulations related to E-waste management.
- **CO3:** Recognize the tools and technologies required for e-waste recycling assessment and monitoring.
- **CO4:** Address contemporary issues and analyse the local and global impact of computing and engineering solutions on individuals, organizations, and society.
- **CO5:** Apply the knowledge about E-waste management in routine daily life to minimize the hazards.

Personal Insight: I find **CO5** to be the most practical and useful, as it empowers us to directly apply these strategies in our day-to-day lives to mitigate environmental risks.

3b. How are these COs mapped to the Program Outcomes (POs)?

The mapping is structured as follows (Weightage format: PO/Max 12). Other POs not listed do not have direct mapping weightages for this specific course.

Course Outcome	PO 1 (Knowledge)	PO 4 (Investigation)	PO 5 (Modern Tools)	PO 6 (Society)	PO 7 (Environment)
CO 1	1/12	-	-	-	1/12
CO 2	-	-	-	2/12	3/12
CO 3	1/12	1/12	-	-	-
CO 4	2/12	1/12	3/12	3/12	-
CO 5	-	2/12	-	-	1/12

Q4. Real-World Applications

4. Where do you see the applications of this course in real life and industry?

- **Real Life:** Safely disposing of personal non-functioning electronic gadgets like old computers, smartphones, TVs, and charging cables.
- **Industry:** Highly applicable in sustainable resource recovery, government environmental monitoring, and E-waste recycling firms. For instance, companies like Eco Recycling Ltd (Ecoreco) in Mumbai directly utilize these principles.

Q5. Global Perspective

5a. Select one Top university outside India and mention the name of their similar course.

Delft University of Technology (TU Delft) in the Netherlands offers a closely related course titled *"Waste Management and Critical Raw Materials"*.

5b. How does their curriculum compare with our course?

Both programs share a core focus on e-waste management and recycling technologies. However, the TU Delft curriculum integrates a heavier emphasis on the broader "circular economy" and highly advanced recovery mechanisms.

Q6. Course Logistics & Resources

6a. How many lecture sessions are planned for this course?

A total of **30 teaching hours** are planned to cover all 6 modules.

6b. Learning Resources & Library Availability

Resource Type	Title & Author	Available in Library?
Textbook	*Environmental Science* by G. Tyler Miller Jr. & Scott Spoolman	Yes
Reference	*Electronic Waste Management* by Hester R.E. & Harrison R.M.	No
Reference	*Electronic Waste* by Fowler B.	No
Reference	*Environmental Science and Engineering* by Keerthinarayana	Yes
E-Book	*Electronic Waste Management Rules 2016* (CPCB)	No
Research Paper	*E-Waste Management: As a Challenge to Public Health in India*	No

6c. What resources are currently uploaded on V-Refer?

The V-Refer portal provides access to comprehensive question banks, PPTs, class test papers, class notes, digital content, and former Internal Assessment (IA) question papers with solutions.

6d. What are the stipulated office hours of faculty for doubt-solving sessions?

For our TE CMPN batch, doubt-solving sessions are held on **Thursdays from 4:00 PM to 5:00 PM** in the D101 faculty room.

Q7. Evaluation & Assessment Scheme

7a. What is the Evaluation Metric for this Theory Course?

This is a purely internal course. The metric is 100% based on the **Internal Semester Assessment (ISA)** which carries a weightage of **75 marks**. Mid-Semester (MSE) and End-Semester (ESE) exams are Not Applicable (NA).

7b & 7c. What does ISA comprise of, and what is its rubric?

The ISA (75 marks total) is continuously evaluated through the following varied tasks:

- Class Participation (5 Marks)
- Online Game-based Competition via Kahoot (5 Marks)
- Video-based Task via Edpuzzle (5 Marks)
- Carbon Footprint Calculator Assessment (5 Marks)
- Device Anatomy & Component Identification (5 Marks)
- Class Test with Peer Assessment (10 Marks)
- Case Studies on Global E-Waste (10 Marks)
- Data Analysis Activity visualizing e-waste generation (10 Marks)
- Sustainability Reel / Podcast creation (10 Marks)
- Campus Sustainability Consultancy & Audit (10 Marks)
- Green Hackathon / Mini Project (10 Marks)

7d. What are the dates of MSE for this course?

While the general institutional schedule may block dates for HSSM exams, the AAP specifically designates MSE as **NA (Not Applicable)** for our evaluation scheme.

7e. What is the date of ESE for this course?

There is **no ESE component** for this subject.

Q8 - Q10. Practical Details & Academic Calendar

8. Is there an allied Practical Course?

No, this subject does not have a separate allied practical course. *(Sub-questions 8b, 8c, 8d are Not Applicable)*.

9. What is the date of declaration of Result?

The final results for the semester are expected to be declared on **19 December 2026**.

10. When does the next Academic Term Start?

The subsequent academic term is slated to begin on **4 January 2027**.